Current Midterm 2023 ZOO603
Subjective part
Composed and compiled by Farhan Ahmed

Fugu/ puffer poisoning marks3

Pufferfish poisoning • Tetrodotoxic poisoning occurs following the ingestion of pufferfish, boxfish or porcupine fish. • This is also known as puffer or fugu poisoning. • The poisoning is due to toxin known as tetrodotoxin which is concentrated in the internal organs such as liver, ovary and intestines. • The skin also contains some glands which secrete these toxins. • Usually the body muscles are free of the toxin.

Lateral line systems marks3

The lateral line system in fish is an important sensory organ. It helps fish detect vibrations and movements in the surrounding water. The system consists of a network of small canals called lateral lines that run along the sides of the fish's body. Inside these canals are sensory cells called neuromasts, which are sensitive to changes in water pressure. The lateral line system plays a crucial role in helping fish navigate, find prey, and avoid predators.

3 features of Chondrichthyes marks3

- **Chondrichthyes is a class of fishes included in the division Gnathostomata because they have jaws.**
- **They possess tail fins, which help with propulsion and maneuvering in water.**
- **Unlike other fish, Chondrichthyes lack swim bladders or lungs, which affects their buoyancy control.**
- **They have a large upper lobe in their tails, providing stability and efficient swimming.**
- **Chondrichthyes have a skeleton made of cartilage rather than bones, known as a cartilage skeleton.**
- **They possess paired appendages, such as pectoral and pelvic fins, which aid in movement and control.**
- **Unlike bony fish, Chondrichthyes do not have an operculum, a bony flap covering the gills.**

General characteristics of fish marks5

- **Fish are aquatic animals that live in water and respire through gills.**
- **Unlike terrestrial animals, fish lack limbs with digits (fingers or toes).**
- **Most fish are cold-blooded, meaning their body temperatures can vary with changes in the surrounding environment.**
- **Their body is divisible into head, trunk and tail. Neck is absent.**
- **It has a spindle shaped body. It is helpful in swimming.**
- **Respiration is by gills. Gills are the extensions of the pharynx.**
- **On the head a pair of nostrils are present internal nostril are absent. In Dipnoi Internal nostrils are present.**
- **In the development of fish's amnion is not developed. Hence these animals are called anamniotes.**

- **On the head a pair of eyes is present.**
- **On the lateral sides of the body LATERAL LINE SENSE organs are present. They detect the pressure changes of water.**
- **The body shows paired and unpaired fins. Pelvic and pectoral fins are paired.**
- **Dorsal and ventral fins are unpaired. They maintain balance in water. They are**
- **useful for locomotion.**
- **The digestive system is well developed. In the intestine of shark scroll valve Is present.**
- **The nervous system contains brain and spinal cord. Brain is small & it will not occupy the entire cranial cavity.**
- **10 pairs of cranial nerves are present.**
- **Kidneys are mesonephrlc.**
- **Urinary bladder is absent**
- **The scales of fish originate from the mesoderm (skin); they may be similar in structure to teeth.**
-

Differentiate between actinopterygii and Sarcopterygii marks5

Subclass Actinopterygii (Ray-finned fish):

- **Their fins lack muscular lobes that is why called as ray-finned fishes.**
- **They typically possess swim bladders, gas-filled sacs along the dorsal wall of the body cavity that help regulate buoyancy.**
- **Paired fins supported by dermal rays**
- **Base of fins is not muscular**
- **Tail fins almost equal to upper and lower lobes**
- **They have Blind olfactory sacs.**
- **Examples of ray-finned fishes include various species such as salmon, trout, perch, and catfish.**

Subclass Sarcopterygii (Lobe-finned fish):

- **Have muscular lobes associated with their fins and usually use lungs for gas exchange.**
- **They possess pneumatic sacs that can function as either lungs or swim bladders.**
- **Sarcopterygii have paired fins with muscular lobes.**
- **Some bony fishes are hermaphrodites and a number of species exhibit parthenogenesis**
- **Fertilization is usually external**
- **Development is usually oviparous (egg laying) but can be ovoviviparous or viviparous**
- **Examples of Sarcopterygii include lungfishes and coelacanths.**

what you know about chondrostean marks5

Chondrosteans are a group of ancient ray-finned fish with a cartilaginous skeleton. They are known for their distinctive ganoid scales, which are thick and diamond-shaped.

Chondrostean are thought to be the most primitive of the ray-finned fishes. • They were at their greatest diversity during the Paleozoic and Mesozoic eras. • Ancestral chondrosteans had a bony skeleton, but living members, the sturgeons and paddlefishes, have cartilaginous skeletons. • Chondrosteans also have a tail with a large upper lobe. • Most sturgeons live in the sea and migrate into rivers to breed. • (Some sturgeons live in freshwater but maintain the migratory habits of their marine relatives.) • They are large (up to 1,000 kg), and bony plates cover the anterior portion of the body. • Heavy scales cover the tail. • The sturgeon mouth is small, and jaws are weak. Sturgeons feed on invertebrates that they stir up from the sea or riverbed using their snouts. • Because sturgeons are valued for their caviar (eggs), they have been severely overfished.

Importance of systematics and taxonomy marks3

Importance of Taxonomy and systematics

➢ Sound systematic data are important for conservation. Without accurate taxonomy, ➢ it is impossible to identify the species and evaluate their conservation status, ➢ it is impossible to properly manage their fisheries, ➢ it is impossible to evaluate ➢ the conservation value of habitats or areas, ➢ it is impossible to establish strategies ➢ it is impossible to set priorities.

Differentiate herbivore, carnivore, and omnivore or Feeding groups of fishes marks5 or 3

Carnivores

• Fish exclusively consume animal matters as food.

Herbivores

• Fish exclusively consume plant matters as food.

Omnivores

• Fish consume both plant and animal matters as food.

• Larvivores

• Fish exclusively consume insect larvae as food.

Planktivorous

• Fish exclusively consume plankton as food.

• Voracious

• Fish consuming or craving large quantities of food.

• Detritovores

• Fish consume decaying plants and animals.

Cannibalistic

• Cannibalism is the act of consuming another individual of the same species as food.

❑ Piscivores

• A piscivore is a carnivorous animal that eats primarily fish.

Discuss detritivore 3

Detritivores

• – Fish consume decaying plants and animals matter as food.

• – 10 percent of all fish species

• – feed on decomposing organic matter

Detritivores may also obtain nutrition by coprophagy, which is a feeding strategy involving the consumption of feces.

• Examples

• – lobsters

• – fiddler crabs

Classification of J.Muller marks5

Muller's classification (1844):

- **J. Muller proposed an early classification system in 1844, dividing class Pisces into six subclasses.**

- **The six subclasses proposed by Muller were: Dipnoi (lungfishes), Teleostei (bony fishes), Ganoidiei (Polypterus, Amia, Lepidosteus), Elasmobranchi (cartilaginous fishes), Marsipobranchi (cyclostomes), and Leptocardii (amphioxus).**

Class of cephalaspidomorphi

Sucking mouth
• Teeth are present
• Rasping tongue
• Seven pairs of slits
• Blind olfactory sacs
• Cephalospidomorph were, like most contemporary fishes, very well armored. They also lacked a swim bladder, and would not have been able to keep afloat without actively swimming.
• The head shield provided some lift though, and would have made the cephalaspidomorphi better swimmers than most of their contemporaries.
• The head shield was particularly well developed, protecting the head, gills and the anterior section of the viscera.
Example
• Lampreys

Classification of fishes on basis of habitat? Marks5

Classification of fishes based on habitat:

1. **Freshwater fish: These fishes inhabit freshwater environments such as rivers, lakes, and streams. Examples include trout, catfish, and carp.**
2. **Marine fish: Marine fishes live in saltwater habitats such as oceans and seas. They are adapted to the saline conditions and include species like tuna, sharks, and clownfish.**
3. **Brackish water fish: Brackish water fish reside in estuaries, where freshwater and saltwater mix. They can tolerate varying levels of salinity and include species like mudskippers and some types of snook.**
4. **Deep-sea fish: These fishes inhabit the abyssal depths of the ocean. They are adapted to extreme pressure, darkness, and low temperatures. Examples include anglerfish, gulper eels, and hatchetfish.**
5. **Coral reef fish: Coral reef fish live in the vibrant and diverse ecosystems of coral reefs. They rely on the reef for food and shelter. Species such as clownfish, butterflyfish, and parrotfish are found in coral reef habitats.**

Write type of scale?

Types of scales in fish
Scales in fish
• Different fishes have different types of scales.
• These different types of scales are made of different types of tissue.
• Types of scales also correspond to evolutionary relationships.
There are four main kinds of scales and numerous variations of each kind.
• Placoid (sharks and rays)
• Cosmoid (lungfishes and some fossil fishes)
• Ganoid (bichirs , Bowfin, paddlefishes, gars, sturgeons)
• Cycloid and Ctenoid (most bony fishes)

Nervous System of fishes.

- **Nervous system**
  As in all vertebrates, the nervous system of fishes is the primary mechanism coordinating body activities, as well as integrating these activities in the appropriate manner with stimuli from the environment.
- **Fish typically have quite small brains relative to body size compared with other vertebrates.**
- **But is some fishes like moemyrids and sharks have large brains.**
- **Brain never fills the entire cavity of the cranium. Remaining space filled with a soft gelatinous mass generally fat.**
- **At the front of olfactory lobes a pair of structure that receive and process signals from the nostrils via two olfactory nerves.**
- **The nervous system in fishes can be divided into two parts: the central nervous system and the peripheral nervous system.**

Central nervous system,
• The central nervous system, consisting of the brain and spinal cord, is the primary integrating mechanism.

peripheral nervous system,
• The peripheral nervous system, consisting of nerves that connect the brain and spinal cord to various body organs, carries sensory information from special receptor organs such as the eyes, internal ears, nares (sense of smell), taste glands.

Significance of habituation

Significance of habituation in fishes:

1. **Adaptation: Habituation helps fishes adjust to their environment by reducing response to non-threatening stimuli.**
2. **Energy conservation: Habituation allows fishes to save energy by minimizing unnecessary responses.**
3. **Predator-prey interactions: Habituation plays a role in predator avoidance by reducing the reaction to predator cues.**
4. **Fishing effectiveness: Habituated fishes are less startled by anglers, increasing the chances of successful catches.**
5. **Conservation: Understanding habituation is crucial for conservation efforts and minimizing human-induced disturbances.**

three characters of chondrocytes

Chondrocytes, the cells found in cartilage tissue, possess the following characteristics:

1. **Produce cartilage matrix.**
2. **Occupy lacunae within the matrix.**

3. **Divide to promote tissue growth and repair.**
4. **Exchange nutrients and waste through diffusion.**

taxonomic characters

Systematic characters, also known as phylogenetic characters, are traits or features that are specifically used to infer evolutionary relationships and construct phylogenetic trees. These characters provide insights into the shared ancestry and evolutionary history of organisms.

- **Morphological features: Physical attributes or structures of an organism.**
- **Genetic sequences: Specific sequences of nucleotides in DNA or RNA.**
- **Behavioral patterns: Observable actions or behaviors exhibited by organisms.**
- **Physiological characteristics: Functional or biological traits and processes of an organism.**

Taxonomic character of fishes

Taxonomic characters:
Taxonomic Characters, used to classify organisms on the basis of similarities and differences.
1. Morphometric characters:
• used for measureable structures
2. Meristic characters:
• used for any countable structures
3. Molecular characters:
• nuclear DNA and Mitochondrial DNA
4. Anatomical character:
• skeleton and characters of the soft anatomy

What kind of food do herbivorous fish eat?

Herbivorous Fish exclusively consume plant matters as food.This type of fish eats aquatic plants, flowers, fruits, seeds, leaves and young stems algae that grow on rocks or pebbles by mouth using wide lips under their snout.

Recommended food for the herbivores would be:

• – Vegetable matter
• – Peas
• – Potatoes
• – Cucumber
• – Sea Food
• – Fresh algae

Bottom feeders (dwellers)

Bottom feeders (dwellers) are fish that inhabit the lower level of the aquarium to feed off the bottom, with their mouth turned down or "under slung" and may have barbs to locate food, such as rotten food. Examples include Barbs, Marigal, Labeo calbasu, Cyprinus carpio, Cirrhina cirrhosus.
However, some fish, like Goldfish, do not show a preference for the level of the aquarium.

What is the feed of bottom feeder?

Bottom feeders eat bottom food such as rotten food.

Bottom feeders (dwellers) primarily eat food that is located at the bottom of the aquatic environment. Their diet typically consists of various types of bottom-dwelling organisms, detritus, algae, decaying matter, and other organic material that settles on the substrate.

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A feeding strategy involving the consumption of feces. coprophagy
Less voracious fish species in Africa Hemichromis
Herbivory is one of the most important processes in maintaining ecological balance on the **Mesoamerican Reef.**
Gnathostome diversity comprises roughly **60,000 species**, which accounts for **99%** of all living vertebrates, including humans.
First jawed fishes appeared in the geologic record about **100 million** years after jawless fish came about.



The number of individuals and total biomass_____at successive trophic levels as organisms _____in size.

elect the correct option

Reloc

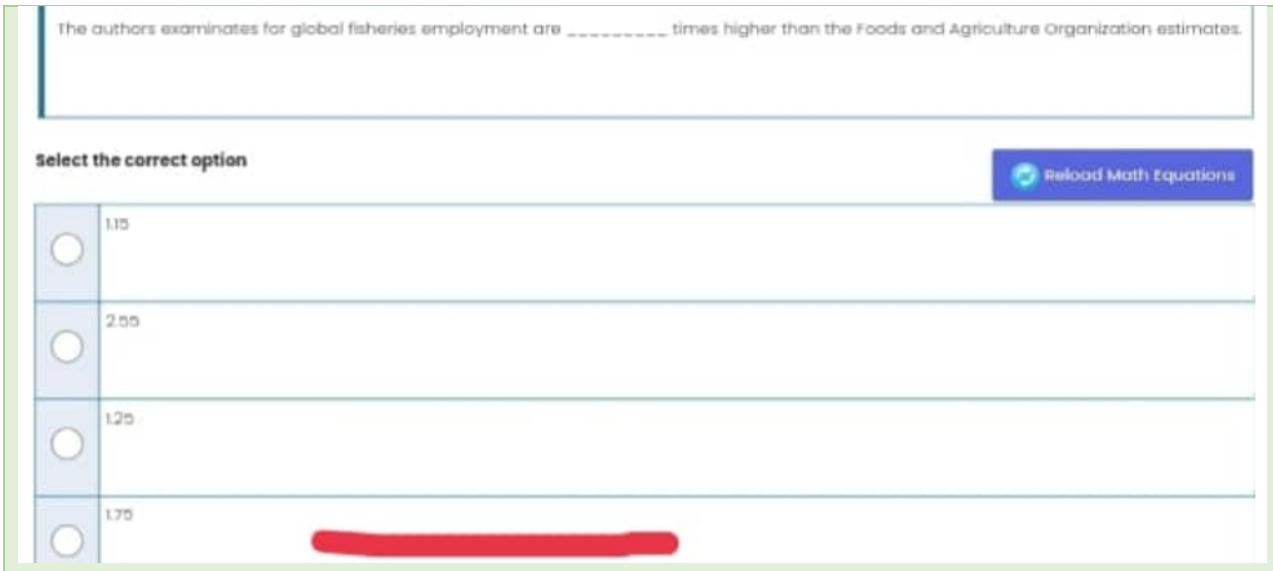
- decrease, increase
- remains same, increase
- none of these
- increase, decrease

The fish flesh is an excellent source of protein, has very little fat, carries a good amount of minerals and vitamins _____ and rich in iodine.

Select the correct option

Reload Math Equation

- A and D
- C and D
- B and D
- A and B



1) ..... is the study of the diversity of fish and the evolutionary Relationships between these organisms. Systematics

2) L.S. Berg classified fishes in to seven classes and separated lampreys and hagfishes into a separate class.

3) Romer's classified fishes into three classes

4) J.S Nelson Classified fishes into five classes and removed class osteichthyes raised sarcoptergii and actinoptergii to class level.

5) Shark is about.....in length. 15m

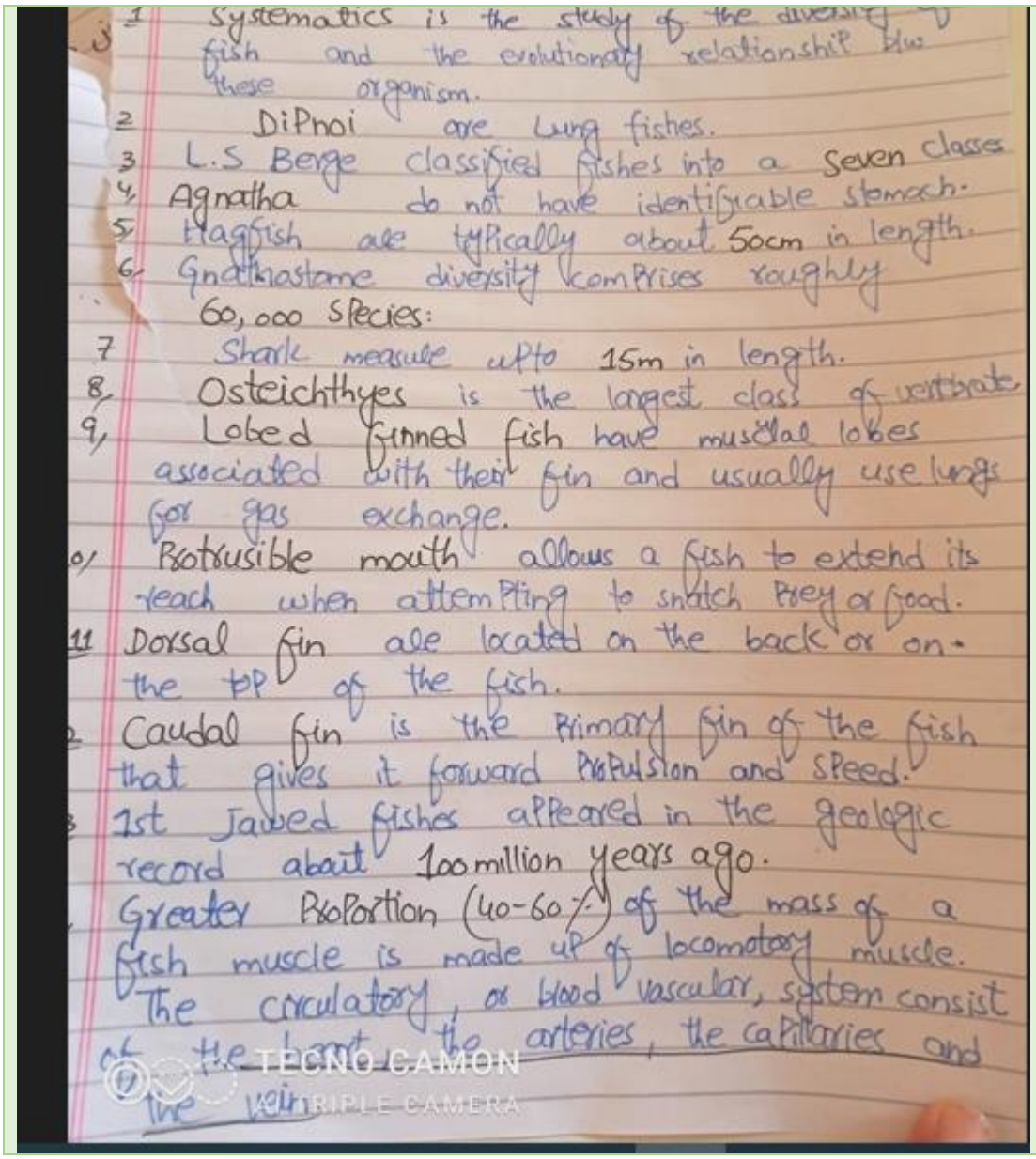
6) Hagfish are typically about 50 cm in length.

7) Hag fishes, belong to Class ...MYXINI

8) Sub class Elasmobranchi contain 700 species

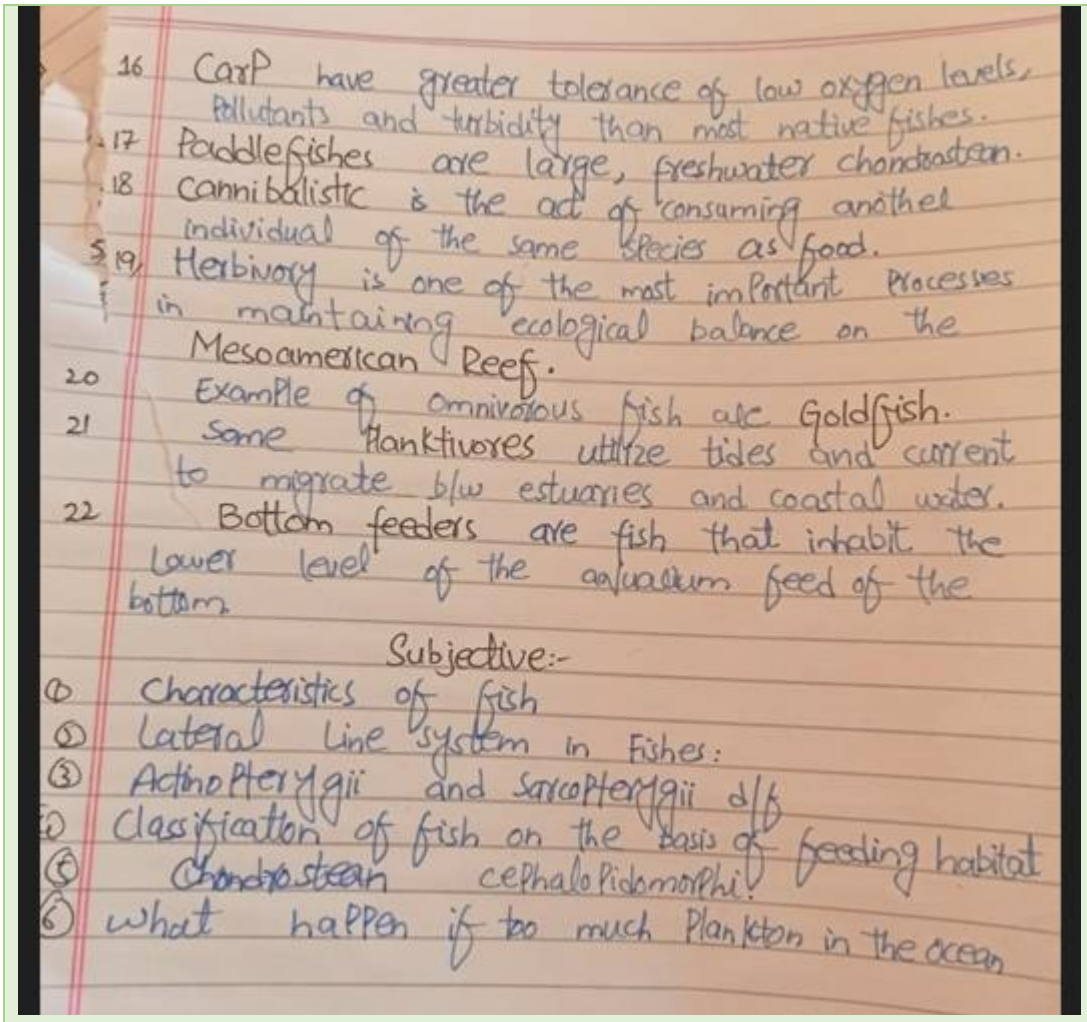
9) Lobe-finned fish Have muscular lobes associated with their fins and usually use lungs for gas exchange.

10) A protrusible mouth allows a fish to extend its Reach when attempting to snatch prey or food particles



1. Systematics is the study of the diversity of fish and the evolutionary relationship btw these organism.
2. DiPnoi are Lung fishes.
3. L.S Berge classified fishes into a seven classes
4. Agnatha do not have identifiable stomach.
5. Hagfish ale typically about 50cm in length.
6. Gnathostome diversity comPrises roughly 60,000 sPecies:
7. Shark measule uPto 15m in length.
8. Osteichthyes is the largest class of vertebrate
9. Lobed finned fish have muscUal lobes associated with their fin and usually use lungs for gas exchange.
10. Protrusible mouth allows a fish to extend its reach when attemPting to snatch Prey or food.
11. Dorsal fin ale located on the back or on the tP of the fish.
12. Caudal fin is the Primary fin of the fish that gives it forward PoPulsion and sPeed.
13. 1st Jawed fishes aPPeared in the geologic record about 100 million years ago.
14. Greater PoPortion (40-60%) of the mass of a fish muscle is made uP of locomotory muscle.
15. The circulatory, or blood vascular, system consist of the heart, the arteries, the caPillaries and the veins

TECNO CAMON
TRIPLE CAMERA


16 Carp have greater tolerance of low oxygen levels, Pollutants and turbidity than most native fishes.
17 Paddlefishes are large, freshwater chondrostean.
18 Cannibalistic is the act of consuming another individual of the same species as food.
19 Herbivory is one of the most important Processes in maintaining ecological balance on the Mesoamerican Reef.
20 Example of omnivorous fish are Goldfish.
21 Some Planktivores utilize tides and current to migrate b/w estuaries and coastal water.
22 Bottom feeders are fish that inhabit the lower level of the aquarium feed of the bottom.

Subjective:-
① Characteristics of fish
② Lateral line system in Fishes:
③ Actinopterygii and Sarcopterygii d/f
④ Classification of fish on the basis of feeding habitat
⑤ Chondrostean cephalopidomorphi
⑥ what happen if too much Plankton in the ocean

Mcqs.
1.omnivore fish... **goldfish**
2.(Systematics) is the study of the diversity of fish and the evolutionary relationships between these organisms.
3. 'Father of Ichthyology..**Peter Artedi**
4. DIPNOI ... **Lung fishes**
5. **L.S. Berg** ...classified fishes in to **seven classes** and separated lampreys and hagfishes into a separate class.
6.**Agnatha**... do not have an identifiable stomach or any appendages.
7. Gnathostome diversity comprises roughly ....**60,000 species**
8. **700 species**..Elasmobranchi
9. **ostichthys** is the largest class of vertebrates.
10.(**Lobe-finned fish**)..Have muscular lobes associated with their fins and usually use lungs for gas exchange.
11. **Beaklike mouth**....used to graze on small algae growing on hard surfaces
12..**Dorsal fins** are located on the back or on the top of the fish
13.. First jawed fishes appeared in the geologic record about ....**100 million years**
14. proportion (**40–60%)..** of the mass of a fish's muscle is made up of locomotory muscle
15. The circulatory, or blood vascular, system consists of ...(**the heart, the arteries, the capillaries, and the veins)**
16. Carp belong to the class… **Actinopterygii- Ray-finned fishes**

🐟🐟MCq 🐟🐟

1) **Systematics** is the study of the diversity of fish and the evolutionary Relationships between these organisms
2) **L.S. Berg** classified fishes in to seven classes and separated lampreys and hagfishes into a separate class.
3) Romer's classified fishes into three classes
4) J.S Nelson Classified fishes into five classes and removed class osteichthyes raised sarcoptergii and actinoptergii to class level.
5) Shark is about.....in length.
6) Hagfish are typically about 50 cm in length.
7) Hag fishes, belong to Class ...MYXINI
8) Sub class Elasmobranchi contain 700 species
9) (Lobe-finned fish) Have muscular lobes associated with their fins and usually use lungs for gas exchange.
10) A protrusible mouth allows a fish to extend its Reach when attempting to snatch prey or food particles
11) Three dorsal fins for locomotion
12) The Caudal fin or Tail fin is the primary fin of the fish That gives it forward propulsion and speed.
13) Placoid scales are found in the sharks and rays.
14) Placoid scales are made of a flattened base with A spine protruding towards the rear of the fish.
15) Lateral line system consist of 100 or more sensory organs
16) Lateral lines in some species contain electroreceptors , which can detect electric Signals in water generated by other organisms.
17) In hagfish proportion (40–60%) of the mass of a fish's muscle is Made up of locomotory muscle.
18) Paddlefishes are large, freshwater chondrosteans.
19) Tetrodotoxic poisoning occurs following the ingestion of pufferfish, boxfish Or porcupine fish.
20) Cannibalism is the act of consuming another individual Of the same species as food.
21) Example of omnivores fish is Goldfish
22) Larvivorous fish are those that feed on immature stages of mosquitoes.

1; __ is the study of diversity of organisms and relationship among them. Systematics ✅
2; Swedish naturalist _ is known as father of icthyology. Peter artedi ✅
3; Subclass Dipnoi cantains.. Lungfish ✅
4; ostiechthyes. Chondrichthyes and placodermi classification was given by. Romer ✅
5; hag fishes are about _ in length. 50cm ✅
6; Gnathostome diversity comprises roughly __, which accounts for 99% of all living vertebrates, including humans. 60,000 species ✅
7; class elasmobranchii contains. 700 species ✅
8; sharks have _ scales. Placoid ✅
9; ___are located on the back or on the top of the fish, and aid the fish in sharp turns or stops, and assist the fish in rolling. Dorsal fins ✅
10; lateral line system consists of _ or more sensory organs. 100 ✅

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11; First jawed fishes appeared in the geologic record about __ years after jawless fish came about. 100 million ✅
12; greater proportion __ of the mass of a fish's muscle is made up of locomotory muscle . (40-60%) ✅
13; __ poisoning occurs following the ingestion of pufferfish, boxfish or porcupine fish. Tetrodotoxic ✅
14; • In advanced teleosts, especially the acanthopterygians, the bladder has lost its connection with the digestive tract, a condition called.. physoclistic ✅
15; fishes contains _ gill opening. 6-7🤔